

EOB Extractor — Data Dictionary

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1. DynamoDB Table: eob-extractions

Access Pattern Summary

Access Pattern	Key Used
Get all extractions for a task	PK = TASK#{taskId}
Get specific extraction	PK = TASK#{taskId}, SK = EOB#{extractionId}
List by status + time	GSI-Status: status = REVIEW_PENDING, sorted by extractedAt
Deduplication check by S3 key	GSI-S3Key: s3Key = {key}
Future: lookup by claim number	GSI-ClaimNumber: claimNumber (currently inert — field not yet populated)
Future: analytics by insurer	GSI-Insurer: insurerName (currently inert — field not yet populated)

Table Configuration

Property	Value
Billing mode	PAY_PER_REQUEST
Partition key	PK (String)
Sort key	SK (String)
Encryption	CUSTOMER_MANAGED — alias/eob-extractor/phi
Point-in-time recovery	Enabled
DynamoDB Streams	NEW_AND_OLD_IMAGES
TTL attribute	ttl (reserved; not currently set by handlers)
Deletion protection	Enabled in production; disabled in sandbox

Item Schema

Attribute	Type	PK/SK/GSI	Required	PHI	Description
PK	String	PK	Yes	No	TASK#{taskId} — partition key
SK	String	SK	Yes	No	EOB#{extractionId} — sort key
extractionId	String	—	Yes	No	ULID generated at entity creation
taskId	String	—	Yes	No	ClickUp task ID from S3 key path

Attribute	Type	PK/SK/GSI	Required	PHI	Description
s3Key	String	GSI-S3Key PK	Yes	No	Full S3 object key (contains taskId, not patient identifiers)
s3VersionId	String	—	No	No	S3 object version ID at time of processing; null if versioning off
status	String	GSI-Status PK	Yes	No	EXTRACTED REVIEW_PENDING FAILED
confidenceScore	Number	—	Yes	No	0.0–1.0; float from Bedrock extraction response
insuranceName	String	—	No	No	Insurance company name (e.g., "Blue Cross Blue Shield of Illinois")
insuranceIdentifier	String	—	No	No	Payer ID or plan number
address	String	—	No	No	Insurance company mailing address (street) — not patient address
city	String	—	No	No	Insurance company city
state	String	—	No	No	Insurance company state abbreviation
zipCode	String	—	No	No	Insurance company ZIP code
locationState	String	—	No	No	State where plan is regulated (may differ from mailing address)
arbitrationPhone	String	—	No	No	Phone for appeals/arbitration
arbitrationFax	String	—	No	No	Fax for appeals/arbitration
arbitrationEmail	String	—	No	No	Email for appeals/arbitration
rawExtractionJson	String	—	Yes	No	JSON-stringified EobExtractionResponse — full Zod-validated output

Attribute	Type	PK/SK/GSI	Required	PHI	Description
modelId	String	—	Yes	No	Bedrock model ID that produced the extraction
classificationResult	String	—	No	No	JSON-stringified classification output from eob-classify-eob
extractedAt	String	GSI-Status SK, GSI-ClaimNumber SK, GSI-Insurer SK	Yes	No	ISO-8601 timestamp of extraction
processingDurationMs	Number	—	Yes	No	End-to-end processing time in milliseconds
correlationId	String	—	Yes	No	ULID; ties all log events for a single execution together
claimNumber	String	GSI-ClaimNumber PK	No	No	Not populated by any handler yet — index is currently inert
insurerName	String	GSI-Insurer PK	No	No	Not populated by any handler yet — index is currently inert
ttl	Number	—	No	No	Unix epoch expiry; reserved for future data lifecycle; currently not set

Global Secondary Indexes

Index	Partition Key	Sort Key	Projection	Purpose
GSI-Status	status (S)	extractedAt (S)	ALL	Query REVIEW_PENDING or FAILED extractions by time
GSI-S3Key	s3Key (S)	—	KEYS_ONLY	O(1) deduplication check before starting new execution
GSI-ClaimNumber	claimNumber (S)	extractedAt (S)	ALL	Future: lookup by claim number (index currently inert)
GSI-Insurer	insurerName (S)	extractedAt (S)	ALL	Future: analytics by insurer (index currently inert)

PHI Assessment — eob-extractions

All fields in this table store **insurance company contact information**, not patient-level identifiers. The `address`, `arbitrationPhone`, `arbitrationFax`, and `arbitrationEmail` fields refer to the insurance company, not to any individual. There is no HIPAA Safe Harbor identifier stored in this table.

No patient PHI is stored in eob-extractions.

2. DynamoDB Table: Insurance-Arbitration-contacts

Status: External, pre-existing table. EOB Extractor reads and conditionally writes. Not CDK-managed by this stack.

Audit finding #007: PITR not confirmed enabled; no audit trail for reads from this table.

Table Configuration

Property	Value
Billing mode	Unknown (pre-existing)
Partition key	Insurance (String)
Encryption	Unknown (pre-existing)
PITR	Not confirmed (Audit #007)

Item Schema

Attribute	Type	PK/GSI	Description
Insurance	String	PK	Composite key: <code>*{locationState} - {insuranceName}</code> (e.g., <code>*IL - Blue Cross Blue Shield</code>)
InsuranceName	String	GSI-InsuranceName-LocationState PK	Plain insurance company name
LocationState	String	GSI-InsuranceName-LocationState SK	State abbreviation where plan is regulated
Address	String	—	Insurance company street address
City	String	—	City
State	String	—	State abbreviation
ZipCode	String	—	ZIP code
ArbitrationPhone	String	—	Arbitration/appeals phone
ArbitrationFax	String	—	Arbitration/appeals fax
ArbitrationEmail	String	—	Arbitration/appeals email

GSI

Index	Partition Key	Sort Key	Used By
GSI-InsuranceName-LocationState	InsuranceName (S)	LocationState (S)	eob-lookup-insurance QueryCommand

Write Behavior (from eob-lookup-insurance)

- `createContact` uses `ConditionExpression: 'attribute_not_exists(Insurance)'` — safe concurrent write guard
- If `ConditionalCheckFailedException` is thrown (concurrent write already created the record), the error is silently swallowed as a no-op
- New contacts trigger a publish to `eob-extractor-review-alerts` SNS topic

Comparison Fields (Mismatch Detection)

When an existing contact is found, these fields are compared (normalized: `.trim().toLowerCase()`):

Extracted Field	Contact Field
address	Address
arbitration_email	ArbitrationEmail
arbitration_fax	ArbitrationFax
arbitration_phone	ArbitrationPhone
city	City
state	State
zip_code	ZipCode

Note: Empty extracted fields with empty contact fields are treated as equal (both sides must be non-empty to compare).

3. S3 Key Structure

Source PDFs — `bucket-specialops` / `bucket-specialops-sandbox`

```
clickup/{taskId}/{filename}.pdf
```

Segment	Description
clickup/	Fixed prefix — required by S3 event notification filter
{taskId}	ClickUp task ID; used as the extraction's <code>taskId</code>
{filename}.pdf	Original filename; never logged (sanitized to <code>clickup/{taskId}/*</code> in logs)

S3 key sanitization: `sanitizeS3Key()` in `audit-logger.ts` strips everything after the last `/`, replacing the filename with `*`. Logged S3 keys are always of the form `clickup/{taskId}/*`.

Quarantine Path (from eob-validate-pdf)

```
quarantine/{taskId}/{filename}.pdf
```

Invalid PDFs are moved to the `quarantine/` prefix instead of being processed.

Audit Logs — `eob-extractor-audit-{account}`

Path	Content
CloudWatch export path	Structured JSON log events exported from <code>/aws/lambda/eob-extractor</code> and <code>/aws/stepfunctions/eob-extractor</code>

4. Zod Schema — Bedrock Extraction Output (`eob-extraction.schema.ts`)

```
z.object({
  is_eob: z.boolean(),
  confidence_score: z.number().min(0).max(1),
  extraction_notes: z.nullable(z.string()),
  insurance_name: z.nullable(z.string()),
  insurance_identifier: z.nullable(z.string()),
  address: z.nullable(z.string()),
  city: z.nullable(z.string()),
  state: z.nullable(z.string()),
  zip_code: z.nullable(z.string()),
  location_state: z.nullable(z.string()),
  arbitration_phone: z.nullable(z.string()),
  arbitration_fax: z.nullable(z.string()),
  arbitration_email: z.nullable(z.string()),
})
```

Schema failures do not abort the pipeline. If Zod parse fails, a zeroed extraction with `confidence_score=0.1` is returned and `isValid=false` is propagated to subsequent steps.

5. PHI Field Blocklist (`audit-logger.ts`)

The following field names are blocked from appearing in any log output. If passed to `logEvent()`, their values are replaced with `[REDACTED]`:

```
patientname, patient_name, memberid, member_id,
subscribername, subscriber_name, ssn, socialsecuritynumber,
social_security_number, dateofbirth, date_of_birth, dob,
address, phonenumber, phone_number, email,
accountnumber, account_number
```

Audit finding #004 (MEDIUM): This blocklist does not cover all 18 HIPAA Safe Harbor identifiers. Missing categories include: fax numbers, medical record numbers, health plan beneficiary numbers, certificate/license numbers, vehicle identifiers, device identifiers, URLs, IP addresses,

and biometric identifiers. Expansion required before any handler change that might introduce new PHI fields.

Note on address : The `address` key is in the blocklist as a precaution. The `address` stored in `eob-extractions` is the insurance company's address, not a patient address. If any future handler passes a field named `address` to `logEvent()` , it will be redacted regardless.